Data Science Capstone Project Report

Predicting SpaceX Falcon 9 First-Stage Landing Success

IBM Applied Data Science Capstone | October 2026

Project repository: https://github.com/namraleuva-lab/spacex-capstone

Executive Summary

- **Goal:** predict whether a Falcon 9 first stage lands successfully, since reuse is what makes SpaceX launches cheaper.
- **Data:** 90 Falcon 9 launches (2010-2020) from the SpaceX REST API and Wikipedia scraping; overall landing success rate 66.7%.
- **Key findings:** success rate climbed from 0% in 2010-2013 to 90% in 2019 and 84% in 2020; KSC LC-39A has the best site record (77.3%); SSO, VLEO, GEO, HEO and ES-L1 orbits succeed most, GTO least among common orbits (52%).
- **Models:** Logistic Regression, SVM, Decision Tree and KNN tuned with GridSearchCV. Logistic Regression, SVM and KNN tie at 83.3% test accuracy; Decision Tree scored 77.8%. SVM was selected on the best cross-validation score (84.8%).

Introduction

- SpaceX advertises Falcon 9 launches at about $62 million, versus $165 million or more from competitors, mainly because the first stage can be reused.
- A company bidding against SpaceX needs to know whether the first stage will land, because that determines the true cost of a launch.
- **Questions:** Which factors (payload mass, orbit, launch site, flight number, booster version) influence landing success? How has the success rate changed over time? Can a machine-learning model predict the outcome reliably?
- **Approach:** data collection, wrangling, SQL and visual EDA, interactive analytics (site maps and Dash), and predictive modeling.

Methodology: Data Collection

- **SpaceX REST API:** GET requests to the launches, rockets, launchpads, payloads and cores endpoints; the JSON responses were normalized into a pandas DataFrame and filtered to Falcon 9 launches.
- **Web scraping:** the Wikipedia 'List of Falcon 9 and Falcon Heavy launches' tables were requested with the requests library and parsed with BeautifulSoup into a second DataFrame.
- **Flow:** API request → JSON → DataFrame → filter Falcon 9 → combine with scraped records → CSV.
- **Result:** 90 Falcon 9 launches with flight number, date, booster, payload mass, orbit, launch site, outcome, grid fins, legs, reuse and coordinates.
- **Notebooks:** https://github.com/namraleuva-lab/spacex-capstone

Methodology: Data Wrangling

- Checked missing values per column; PayloadMass nulls were replaced with the column mean, and LandingPad is null where no pad was used.
- Counted launches per site, occurrences of each orbit type, and the frequency of each landing outcome.
- Created the target label **Class** from the Outcome column: True ASDS, True RTLS and True Ocean map to 1 (success); False outcomes and None map to 0.
- Resulting dataset: 90 rows, 18 columns; success rate 66.7% (60 of 90 landings).
- Categorical variables (Orbit, LaunchSite, LandingPad, Serial) were one-hot encoded, giving 80 features for modeling.

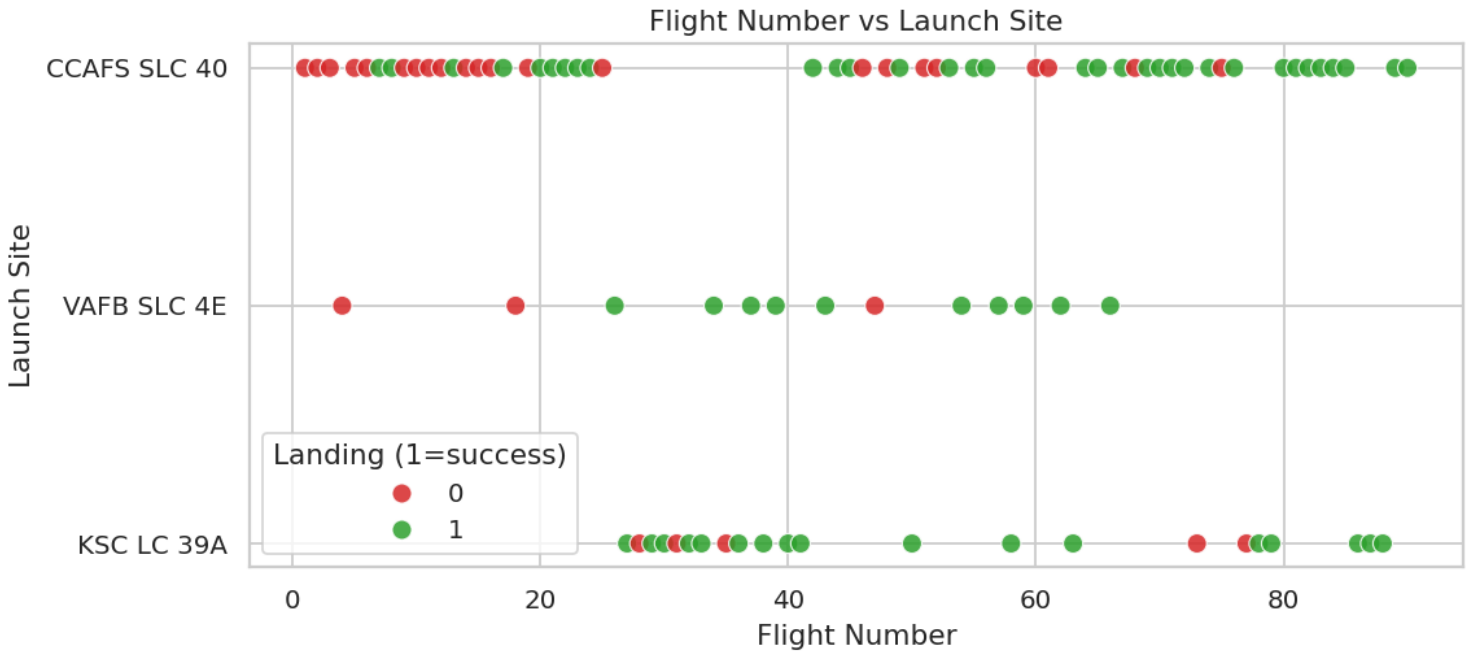
Methodology: EDA and Interactive Analytics

- **Visual EDA (seaborn/matplotlib):** scatter plots of Flight Number and Payload against Launch Site and Orbit, a bar chart of success rate by orbit, and a line chart of the yearly success rate.
- **SQL EDA (SQLite):** ten queries on sites, payload mass, landing milestones, outcome counts, and 2015 failures.
- **Site map:** launch-site locations, success/failure markers per launch, and proximity of sites to the coast and nearest city.
- **Plotly Dash:** launch-site dropdown, success pie chart, payload range slider, and payload vs outcome scatter coloured by booster version.

Methodology: Predictive Analysis

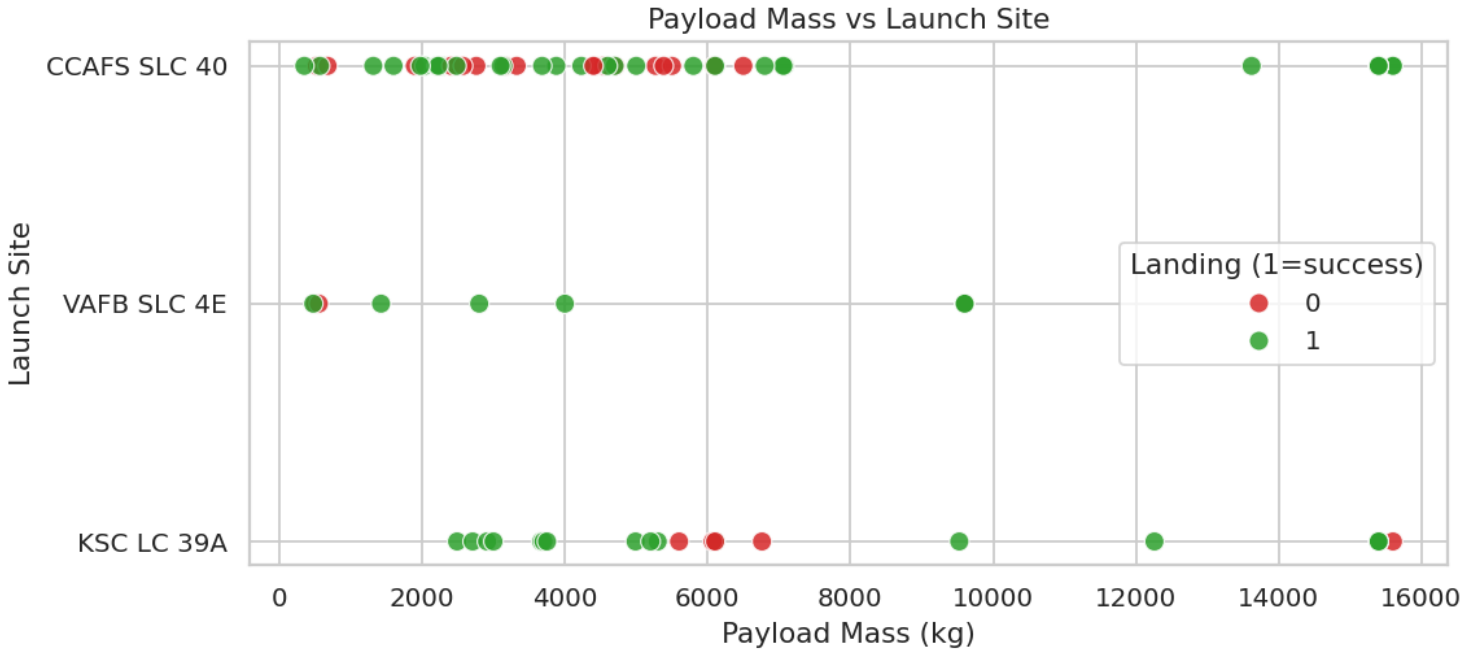
- Features standardized with StandardScaler; data split 80/20 into 72 training and 18 test launches (random_state=2).
- Four classifiers tuned with GridSearchCV using 10-fold cross-validation: Logistic Regression, SVM, Decision Tree, KNN.
- Models were compared on test accuracy; the best model is shown with its confusion matrix.
- Best parameters: SVM kernel=sigmoid, C=1, gamma=0.032; KNN k=6, p=1; Decision Tree max_depth=6, splitter=random; Logistic Regression C=1.

EDA: Flight Number vs Launch Site



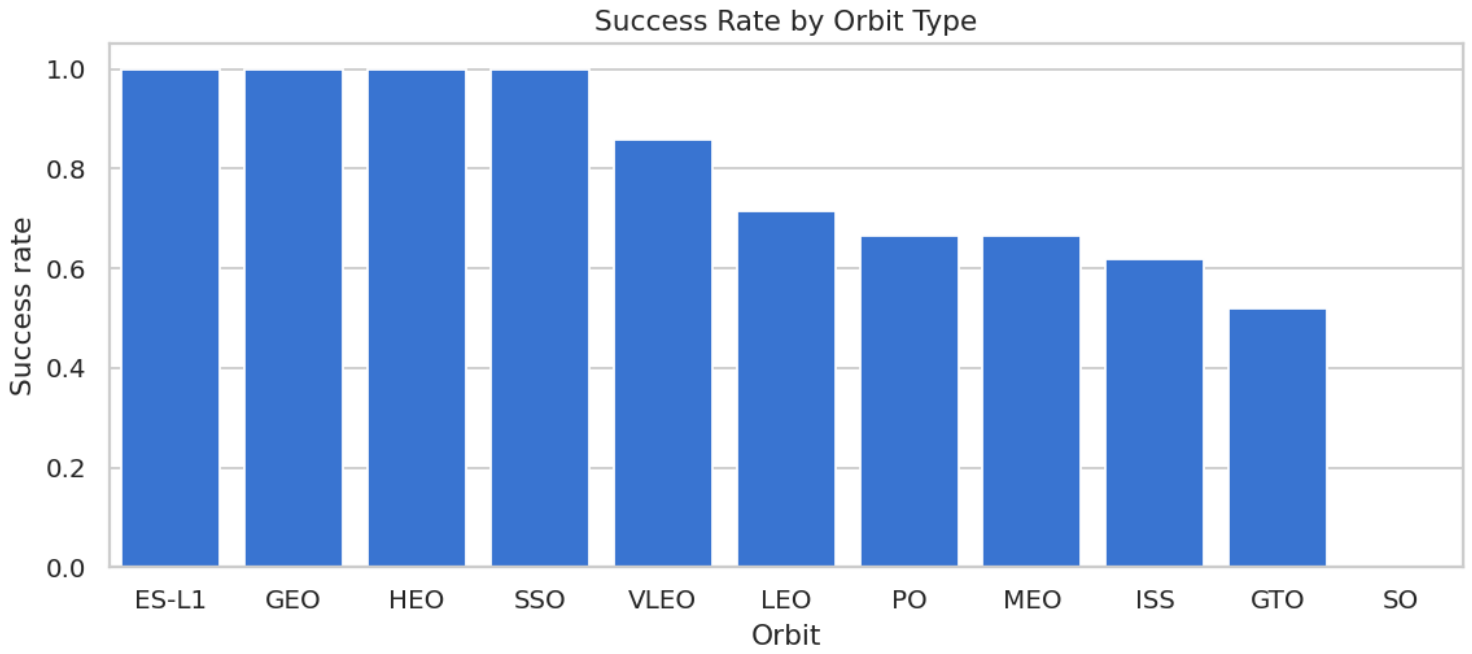
- Success is rare in early flights and common in later ones at every site.
- Flights below 30 succeed 41% of the time; flights 50 and above succeed 80%.
- KSC LC-39A only appears from flight 27 onward, and has a high success rate.

EDA: Payload Mass vs Launch Site



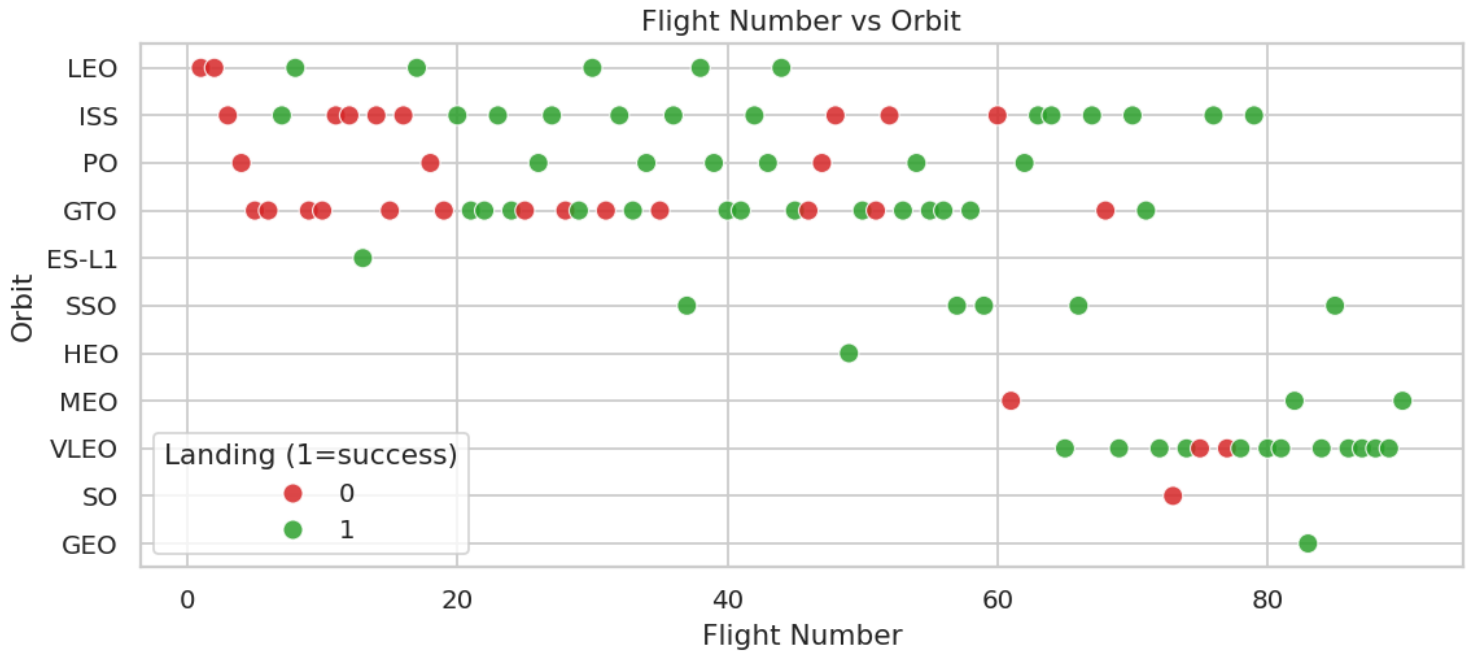
- VAFB SLC-4E has no launches above 10,000 kg (maximum 9,600 kg).
- Of the 15 launches above 10,000 kg (CCAFS and KSC), 13 landed successfully (89% at CCAFS, 83% at KSC).
- Heavy payload alone does not prevent landing.

EDA: Success Rate by Orbit Type



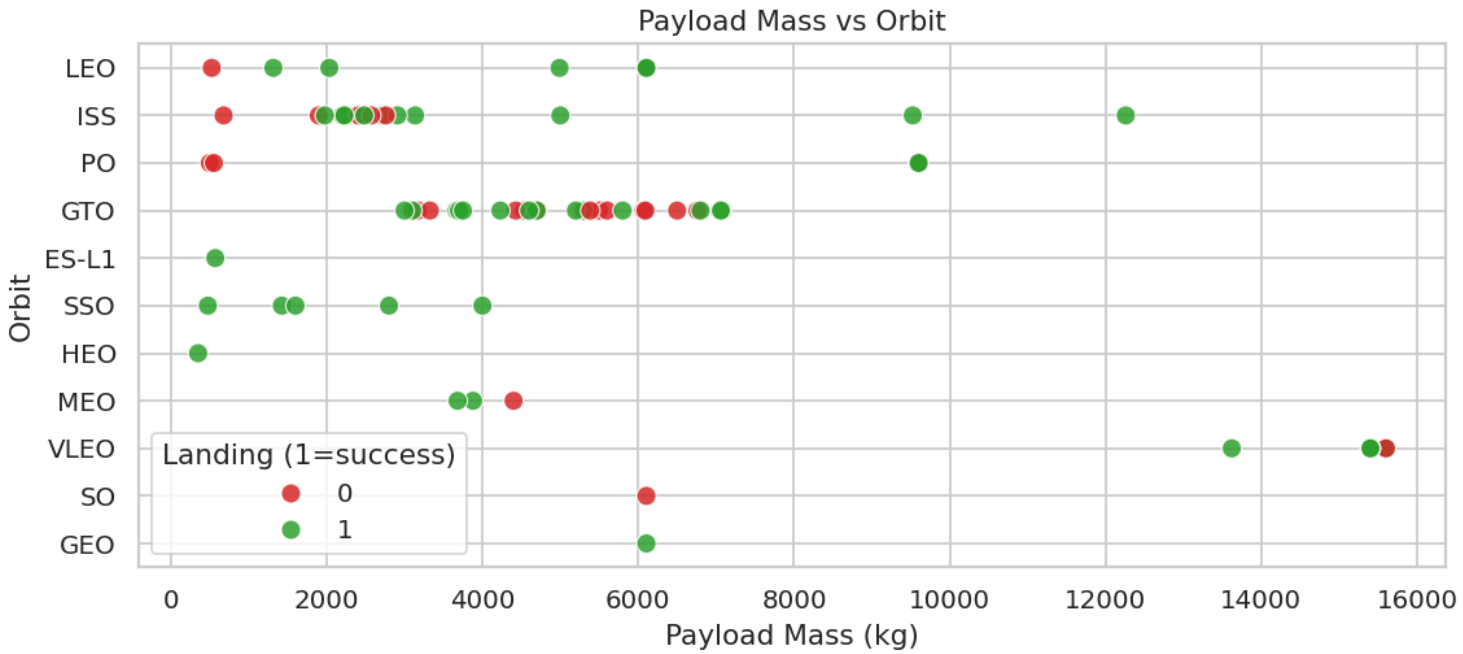
- ES-L1, GEO, HEO and SSO have a 100% success rate, but each has only 1 to 5 launches.
- Among well-represented orbits, VLEO (86%) is highest and GTO (52%) is lowest.
- ISS 62%, LEO 71%, PO 67%; SO has 0% from a single launch.

EDA: Flight Number vs Orbit



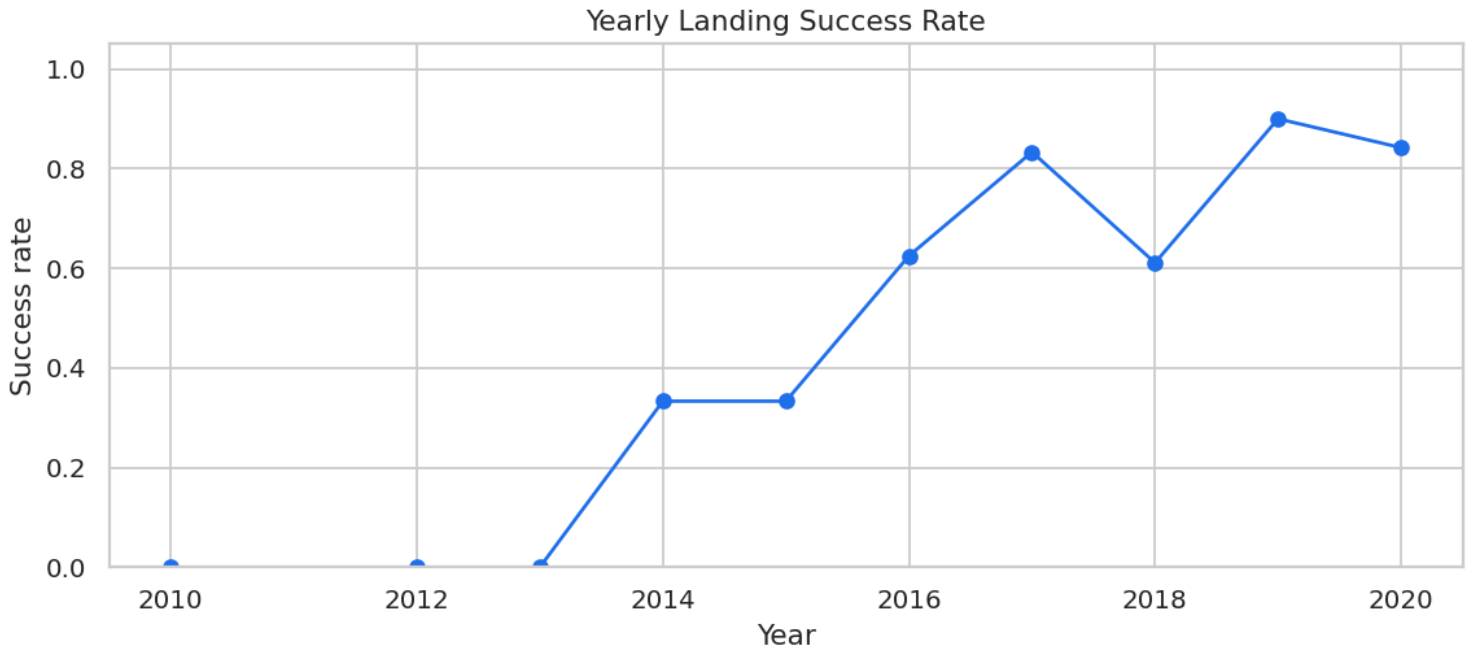
- LEO landings improve with flight number, and VLEO appears only in recent flights (65 and later) with high success.
- GTO shows mixed outcomes across all flight numbers.
- In LEO, success is tied to later flights; in GTO it is not.

EDA: Payload Mass vs Orbit



- VLEO carries the heaviest payloads (about 13,600 to 15,600 kg), yet 86% land.
- GTO payloads are 3,000 to 7,100 kg and show the weakest success.
- ISS and PO have a wide payload range with mixed results.

EDA: Yearly Landing Success Trend



- 0% success in 2010, 2012 and 2013, and 33% in 2014 and 2015.
- Rising trend: 63% in 2016, 83% in 2017, a dip to 61% in 2018, then 90% in 2019 and 84% in 2020.
- SpaceX's landing technology and experience improved over time.

SQL EDA: Launch Sites and Sample Records

Query: `SELECT DISTINCT LaunchSite FROM SPACEX`

LaunchSite
CCAFS SLC 40
VAFB SLC 4E
KSC LC 39A

Insight: Three launch sites are used: CCAFS SLC 40, VAFB SLC 4E (California) and KSC LC 39A.

Query: `SELECT FlightNumber, Date, LaunchSite, Orbit, Outcome FROM SPACEX WHERE LaunchSite LIKE 'CCA%' LIMIT 5`

FlightNumber	Date	LaunchSite	Orbit	Outcome
1	2010-06-04	CCAFS SLC 40	LEO	None None
2	2012-05-22	CCAFS SLC 40	LEO	None None
3	2013-03-01	CCAFS SLC 40	ISS	None None
5	2013-12-03	CCAFS SLC 40	GTO	None None
6	2014-01-06	CCAFS SLC 40	GTO	None None

Insight: The earliest launches from Cape Canaveral (CCAFS) had no landing attempt (Outcome None).

SQL EDA: Payload Mass

Query: `SELECT SUM(PayloadMass) AS total_payload_ISS_kg FROM SPACEX WHERE Orbit='ISS'`

total_payload_ISS_kg
68878.7

Insight: Launches to the ISS (NASA-type missions) carried about 68,879 kg in total.

Query: `SELECT AVG(Payload_Mass_kg) AS avg_payload_F9_v1_1 FROM DASH WHERE Booster_Version LIKE 'F9 v1.1%'`

avg_payload_F9_v1_1
2534.67

Insight: F9 v1.1 boosters carried an average of about 2,535 kg.

Note: the capstone launch table has no customer column, so total payload is computed for ISS-orbit missions in place of the NASA (CRS) filter.

SQL EDA: Landing Milestones

Query: `SELECT MIN(Date) AS first_ground_pad_success FROM SPACEX WHERE Outcome='True RTLS'`

first_ground_pad_success
2015-12-22

Insight: The first successful ground-pad landing was on 2015-12-22.

Query: `SELECT Serial, PayloadMass FROM SPACEX WHERE Outcome='True ASDS' AND PayloadMass BETWEEN 4000 AND 6000`

Serial	PayloadMass
B1022	4696.0
B1026	4600.0
B1021	5300.0
B1031	5200.0
B1046	5800.0
B1046	4000.0
B1059	5000.0

Insight: Six boosters landed on a drone ship with payloads between 4,000 and 6,000 kg: B1021, B1022, B1026, B1031, B1046 and B1059.

SQL EDA: Outcomes and Maximum Payload

Query: `SELECT Outcome, COUNT(*) AS n FROM SPACEX GROUP BY Outcome ORDER BY n DESC`

Outcome	n
True ASDS	41
None None	19
True RTLS	14
False ASDS	6
True Ocean	5
None ASDS	2
False Ocean	2
False RTLS	1

Insight: 41 drone-ship, 14 ground-pad and 5 ocean landings succeeded; 6 drone-ship, 2 ocean and 1 ground-pad attempts failed.

Query: `SELECT DISTINCT Serial, PayloadMass FROM SPACEX WHERE PayloadMass=(SELECT MAX(PayloadMass) FROM SPACEX)`

Serial	PayloadMass
B1048	15600.0
B1051	15600.0

Insight: Boosters B1048 and B1051 carried the maximum payload of 15,600 kg.

SQL EDA: 2015 Failures and Outcome Ranking

Query: `SELECT substr(Date,6,2) AS month, Serial, LaunchSite FROM SPACEX WHERE Outcome LIKE 'False ASDS' AND substr(Date,1,4)='2015'`

month	Serial	LaunchSite
01	B1012	CCAFS SLC 40
04	B1015	CCAFS SLC 40

Insight: Two drone-ship landings failed in 2015: January (B1012) and April (B1015), both from CCAFS SLC 40.

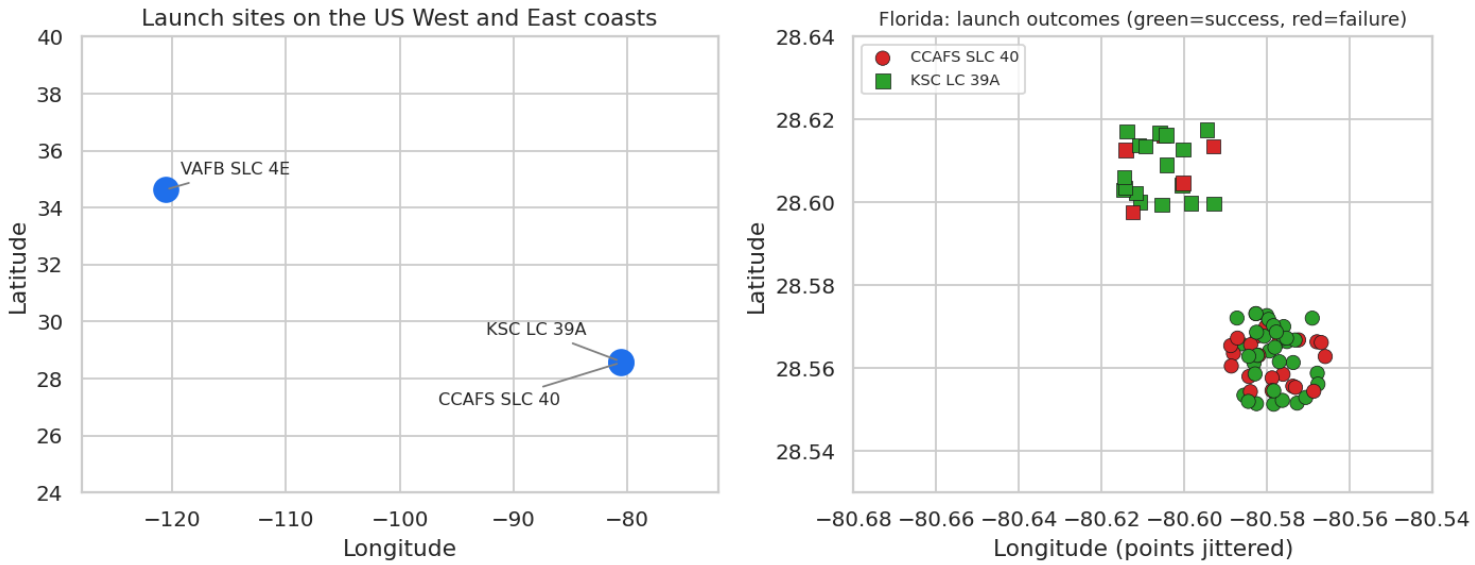
Query: `SELECT Outcome, COUNT(*) AS n FROM SPACEX WHERE Date BETWEEN '2010-06-04' AND '2017-03-20' GROUP BY Outcome ORDER BY n DESC`

Outcome	n
None None	9
True ASDS	5
False ASDS	4
True RTLS	3
True Ocean	3
None ASDS	2
False Ocean	2

Insight: Between 2010-06-04 and 2017-03-20, drone-ship successes (5) slightly outnumbered failures (4); 9 launches had no landing attempt.

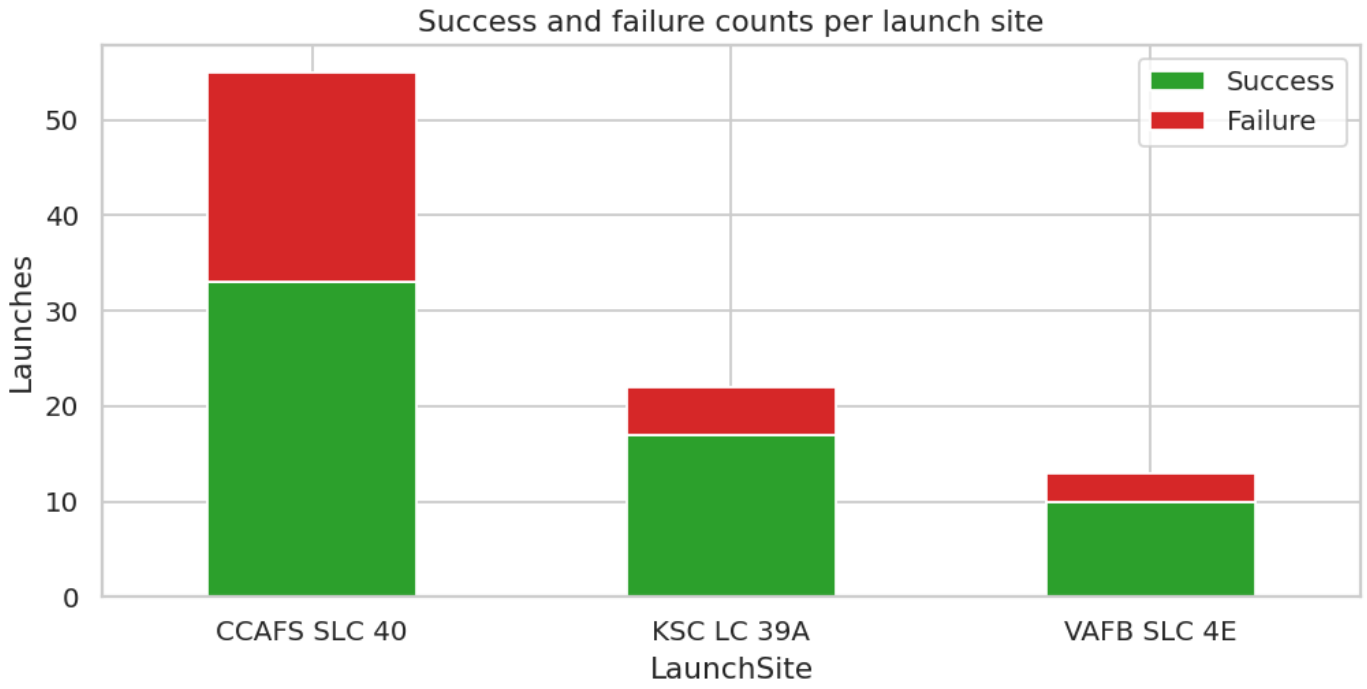
Site Map: Launch Site Locations

Interactive Map



- Three sites: CCAFS SLC-40 and KSC LC-39A on Florida's Atlantic coast, and VAFB SLC-4E on California's Pacific coast.
- All sites are on the coast, so rockets launch over the ocean.
- Right panel: green = successful landing, red = failure for each launch.

Site Map: Success and Failure per Site



- CCAFS SLC-40: 60% success (33 of 55).
- KSC LC-39A: 77.3% (17 of 22), the best record.
- VAFB SLC-4E: 76.9% (10 of 13).
- Maps here are static plots of the site coordinates (no map tiles).

Site Map: Proximity Analysis

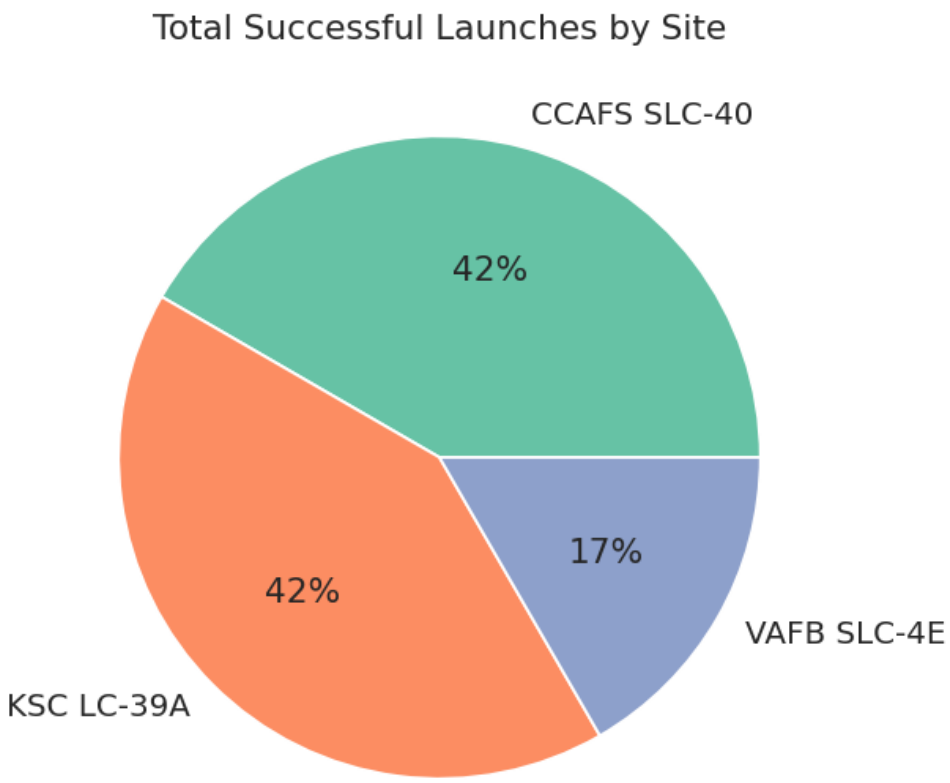
Interactive Map

Launch site	Latitude	Longitude	Nearest city (approx.)	Distance (km)
CCAFS SLC 40	28.5619	-80.5774	Cocoa Beach	27.1
KSC LC 39A	28.6081	-80.604	Titusville	19.9
VAFB SLC 4E	34.6321	-120.6108	Lompoc	14.0

- **Coast:** the site coordinates sit on the shoreline of the Atlantic (Florida) and Pacific (California), so launches go out over water.
- **Cities:** each site is roughly 14 to 27 km from the nearest town (straight-line distance from coordinates, approximate).
- **Conclusion:** sites are near the coast and away from cities so failed launches and spent stages fall in the ocean rather than on populated areas, and are close to transport links for equipment.
- Railway and highway distances are not included: they require measuring on an interactive map.

Dashboard: Successful Launches by Site

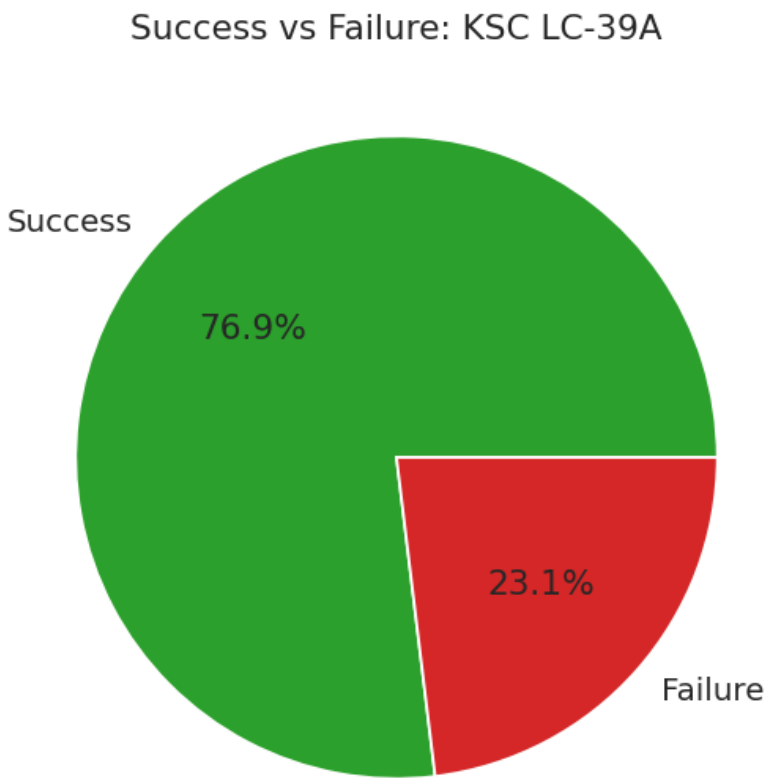
Plotly Dash



- KSC LC-39A has the largest share of successes (10 of the 24 successful launches in the dashboard dataset).
- CCAFS SLC-40 also has 10 successes, but from 33 launches (30% success).
- VAFB SLC-4E has 4 successes (40%).

Dashboard: Highest Success-Rate Site

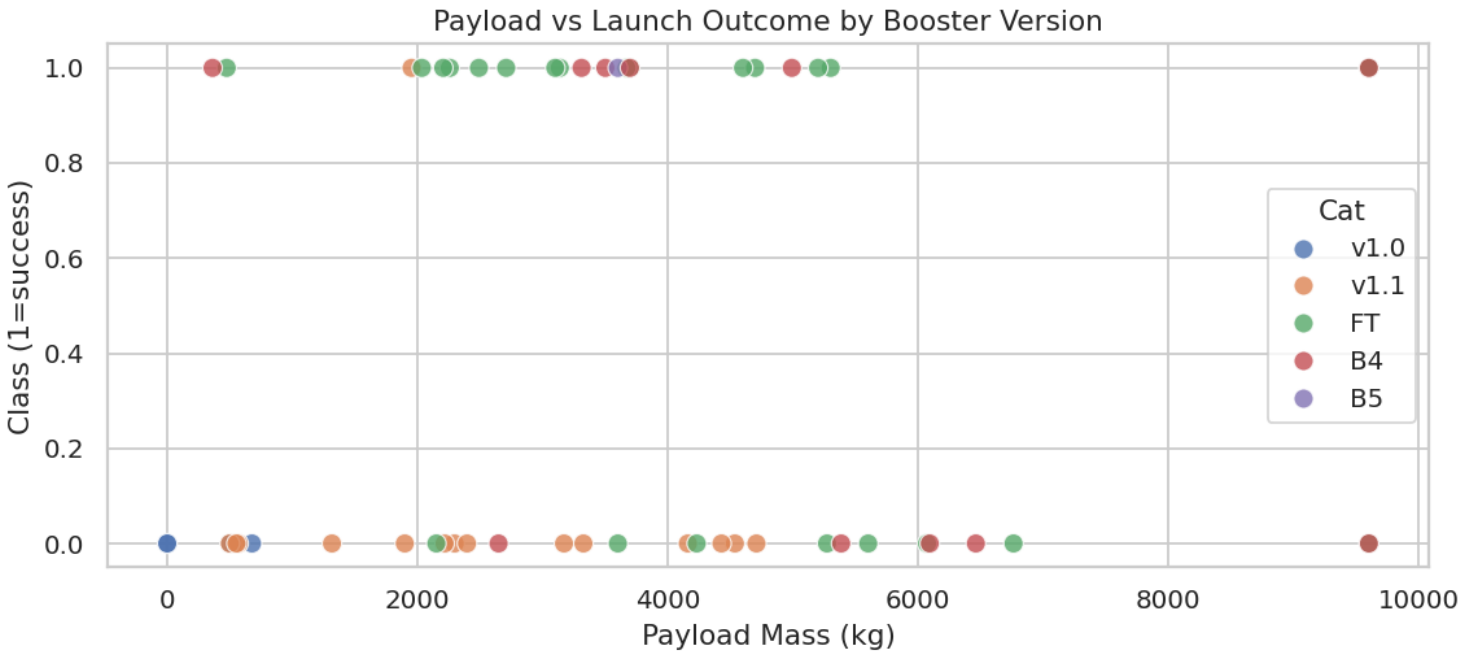
Plotly Dash



- KSC LC-39A has the highest success rate: 10 successes and 3 failures (76.9%).
- This matches the full 90-launch dataset, where it also leads.

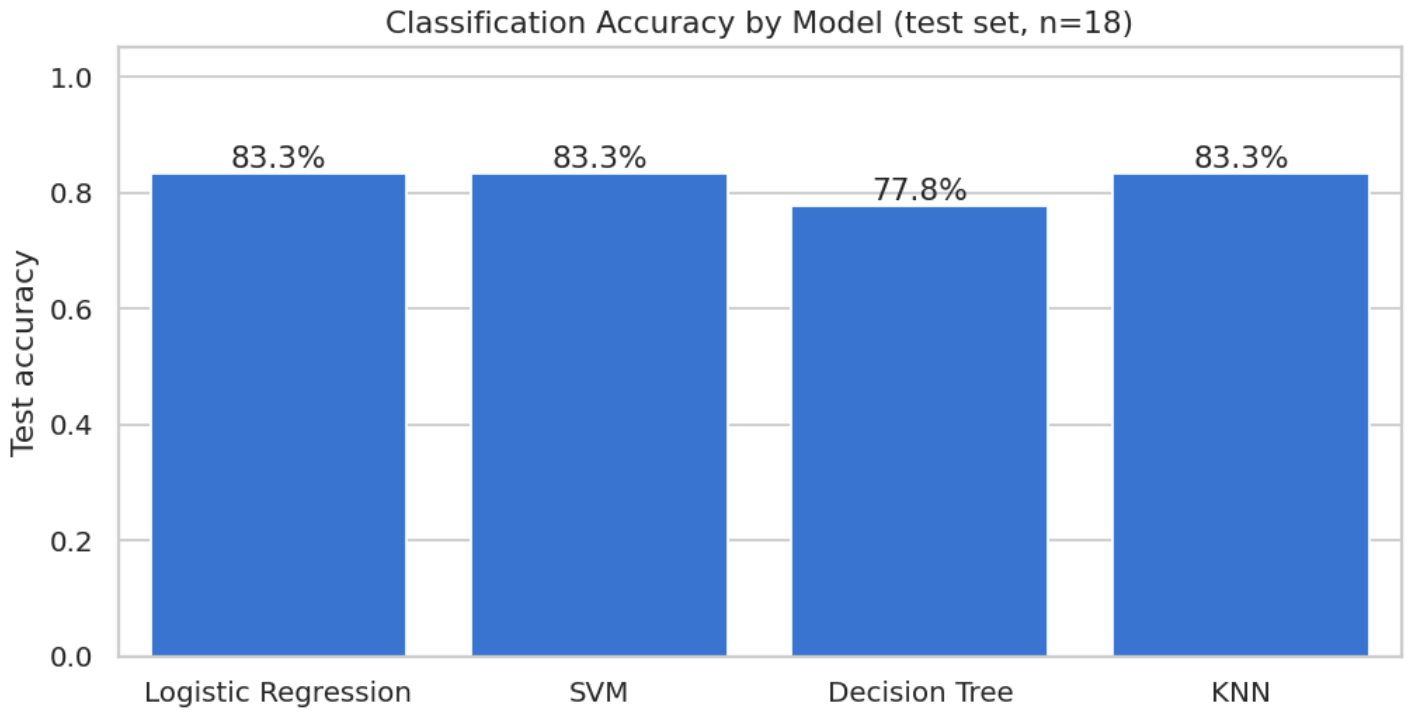
Dashboard: Payload vs Launch Outcome

Plotly Dash



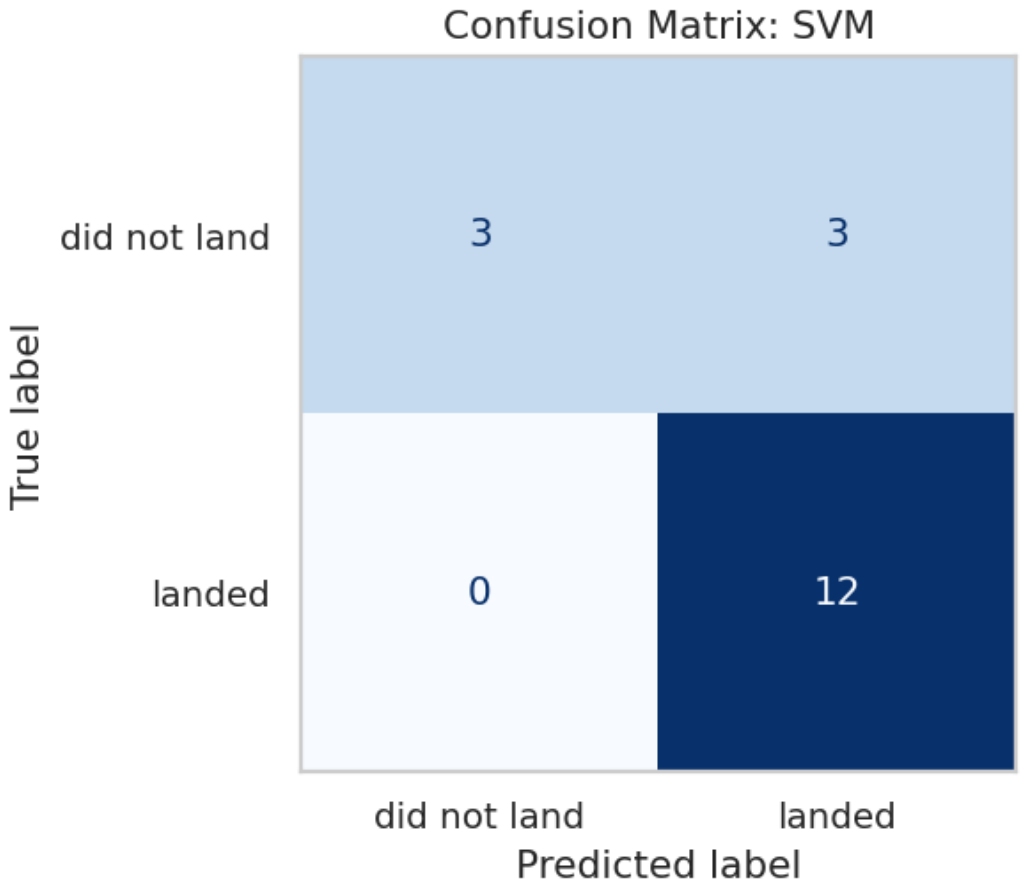
- By payload range, success is 23% for 0-2,000 kg, 62% for 2,000-4,000 kg, 38% for 4,000-6,000 kg and 33% above 6,000 kg.
- Booster version matters most: v1.0 and v1.1 almost never landed, FT succeeded 67%, and B4 and B5 are the latest.
- The dashboard slider lets you filter by payload range and site.

Predictive Analysis: Classification Accuracy



- Logistic Regression, SVM and KNN each reach 83.3% test accuracy (15 of 18); Decision Tree reaches 77.8%.
- Cross-validation scores: SVM 84.8%, Decision Tree 83.6%, KNN 83.4%, Logistic Regression 82.1%.
- SVM was chosen as the best model on cross-validation score, but differences between models are small on only 18 test samples.

Predictive Analysis: Confusion Matrix (SVM)



- True negatives: 3, false positives: 3, false negatives: 0, true positives: 12.
- The model found every successful landing (no false negatives) but predicted 'landed' for 3 of the 6 launches that did not land.
- The main weakness is false positives: it is optimistic about landings.

Conclusions

- **Success improved over time:** from 0% (2010-2013) to about 90% (2019-2020), so experience is a strong factor.
- **Orbit and payload:** SSO, VLEO and other small-sample orbits succeed most, GTO least (52%); heavy payloads above 10,000 kg still land 89% (CCAFS) and 83% (KSC) of the time.
- **Site:** KSC LC-39A (77%) and VAFB SLC-4E (77%) outperform CCAFS SLC-40 (60%).
- **Model:** SVM, Logistic Regression and KNN reach 83.3% test accuracy; SVM chosen on cross-validation (84.8%). It misses no successful landings but has false positives.
- **Business takeaway:** a competitor should assume recent Falcon 9 launches usually land and price a bid accordingly, but treat GTO missions as higher risk.
- **Limitations:** only 90 launches and an 18-launch test set, so results vary with the train/test split and model differences are not statistically significant.

Appendix

- **GitHub:** https://github.com/namraleuva-lab/spacex-capstone
- **Dash app:** spacex_dash_app.py (dropdown, pie chart, payload RangeSlider, scatter chart).
- **GridSearchCV:** `GridSearchCV(estimator, parameters, cv=10).fit(X_train, Y_train)`
- **Class label:** `landing_class = [0 if o in bad_outcomes else 1 for o in df['Outcome']]`
- **API call:** `requests.get('https://api.spacexdata.com/v4/launches/past')`
- **Data note:** SQL queries ran in SQLite on the capstone launch table; datasets are the course's standard dataset_part_2 / dataset_part_3 and spacex_launch_dash files.